BMC Public Health

RESEARCH

Open Access

Knowledge, practices and barriers to sun protection among farmers in Bagua Grande (Peru): a mixed-methods study

Nelson Cesar Santisteban Salazar[1*], Mirtha Yvis Santisteban Salazar[2] and Magnolia Anacarina Arrasco Barrenechea[3]

Abstract

Background Farmers are at risk of excessive exposure to solar radiation. Knowledge and practices in photoprotection were evaluated and barriers to sun protection were explored among insured farmers in Bagua Grande.

Materials and methods A mixed methodology with a sequential explanatory design with quantitative dominance was used. A questionnaire in Spanish, "CHACES," was administered to 203 farmers selected by simple random sampling, and seven people were interviewed in depth to achieve data saturation. The survey data were synthesized using Stata statistical software, and the interviews were analyzed thematically.

Results 83% of respondents had low knowledge of photoprotection. Farmers are exposed to the sun during midday hours (62%), do not use sunscreen (53%) or sunglasses (88%), although they do wear long sleeves and pants (42%). Significant differences in knowledge levels were found with age ($p < 0.001$), natural skin color ($p < 0.001$), and use of sunscreen ($p = 0.002$). The most common barriers were lack of information about sun protection measures, discomfort from wearing protective clothing, the high cost of sunscreen, and access to health services.

Conclusion Farmers' knowledge of sun protection was low and practices were inadequate, and the barriers identified prevent the practice of sun protection. Therefore, interventions aimed at modifying behavior and practices in photoprotection according to the characteristics of the population are required, using behavioral measures and barrier methods to safeguard both their health and safety.

Keywords Farmers, Knowledge, Habits, Sunlight, Ultraviolet rays

Introduction

Solar radiation interacts with biological tissues primarily through photochemical mechanisms associated with ultraviolet (UV) radiation, which is responsible for most adverse health effects [1]. In recent decades, the incidence of skin cancer has increased by 3–7% among fair-skinned populations, particularly in North America, Europe, and Oceania [2]. The most frequent types are basal cell and squamous cell carcinomas, both non-melanoma and predominant in older adults, whereas melanoma, although less common, accounts for nearly 90% of skin cancer deaths due to its high metastatic potential [3, 4], representing a major global health and economic burden [5].

While moderate sun exposure provides benefits such as vitamin D synthesis and the regulation of biological functions [6–8], overexposure leads to harmful effects, including skin cancer [9], photoaging, and ocular and

*Correspondence:
Nelson Cesar Santisteban Salazar
nelson.santisteban@essalud.gob.pe
[1]Seguro Social de Salud, Lima, Peru
[2]Universidad Cesar Vallejo, Chiclayo, Peru
[3]Universidad Ricardo Palma, Lima, Peru

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

immunological alterations [10, 11]. Risk increases in photoexposed areas and among individuals with low skin phototypes [12], although people with darker skin are not exempt [13]. Key risk factors include age, gender, education, occupation, clothing, pigmentation, and family history [14–19], all of which are directly linked to sun-exposure habits [20].

Outdoor workers, particularly farmers, experience cumulative occupational sun exposure [1], which is six to eight times higher than that of indoor workers [21, 22]. The lack of effective occupational safety measures [1, 17, 23], along with barriers such as misinformation [24, 25], discomfort, or the impracticality of certain protective behaviors [26], limits the use of photoprotection strategies. Previous studies have shown insufficient knowledge and inadequate sun-protection practices in this group [9, 24, 26], as well as a possible association between the two variables [5, 25, 27]. However, greater knowledge does not always translate into protective behaviors [28, 29], suggesting that mixed-methods approaches may help to better understand this gap.

The risk of sun damage also depends on skin phototype, based on skin color and susceptibility to burning and tanning, as determined by the Fitzpatrick classification [30], and on the ultraviolet index (UVI), adopted by the World Health Organization to measure the potential for sun damage on a scale of 1 to 11+ [31]. Higher UVI values indicate greater risk of skin and eye injury [1, 32]. Peru is one of the countries with the highest solar radiation due to its proximity to the equator [33, 34], with UVI levels ranging from high to extremely high, particularly in the Amazon region [34]. In 2017, skin cancer ranked third among the most frequent neoplasms in the country (10.4%) (787/7537) [35], and in the Amazonas region, it was the most common, accounting for 21.6% (223/1031) of registered cases, mainly in men (51.1%, 114/223) [36].

Bagua Grande, located in the Amazonas region, experiences high levels of solar radiation year-round due to its proximity to the equator [34] and has an economy predominantly based on agriculture [37]. In 2023, the El Buen Samaritano Hospital in this city reported 36 cases of skin carcinoma, 25% of which involved individuals covered by the agrarian insurance system. Although the frequency is moderate, skin cancer has high global mortality, with an estimated one in four deaths occurring in the agricultural sector [38, 39]. This highlights the need for preventive and educational strategies targeting this population.

Although agriculture is one of Peru's main economic activities [40], and more than 50% of the economically active population in Amazonas works in this sector [41], farmers—who are highly exposed to the sun—have received little attention in national sun-safety awareness campaigns, which are mostly directed at urban areas. In this context, the present study focuses on farmers in Bagua Grande affiliated with the agrarian insurance program, aiming to assess their knowledge, practices, and barriers regarding sun safety. Understanding these factors will enable the design of tailored and culturally relevant interventions to promote sustainable photoprotective behaviors among rural populations.

Materials and methods

Study type and design

A mixed-methods study with an explanatory sequential design and quantitative dominance was conducted. The first quantitative phase consisted of an analytical observational design to assess the level of knowledge and sun-protection practices among agrarian insurance holders enrolled at El Buen Samaritano Hospital in Bagua Grande. Based on these findings, a qualitative research phase was initiated to explain or support the results of the first phase. For the qualitative analysis, a phenomenological design was used, as the aim was to understand barriers to sun-safety practices from the experiences of farmers through semi-structured interviews.

Population and sample

The study population consisted of agrarian insurance beneficiaries enrolled at El Buen Samaritano Hospital in Bagua Grande, Peru. The total insured population over 18 years old is 6,128; however, 646 individuals were identified as agricultural workers, according to the local insurance office for the year 2023. From this population, we estimated an expected proportion of 25% adequate knowledge, a 95% confidence level, and a 5% margin of error, obtaining a minimum sample size of 200 participants.

From the complete list of 646 agrarian workers registered in EsSalud, random selection was performed, leaving a final sample of 203 workers using simple random sampling.

Inclusion criteria were: being an agrarian insurance holder (titular or dependent), being older than 18 years, currently working in agricultural activities, being literate, and being able to complete the questionnaire or interview in full. Farmers who did not provide informed consent were excluded.

For the qualitative phase, telephone calls and contact attempts were made with a list of 16 survey participants who had agreed to an interview, provided by the principal investigator. Saturation of information was reached with seven farmers (four women and three men, aged 28 to 60 years).

Instrument

The Spanish-language questionnaire on habits, attitudes, and knowledge related to sun exposure in adults

("CHACES") was used. This questionnaire [42] has psychometric properties and Cronbach's alpha validation values > 0.70; likewise, the behavior and knowledge items showed intraclass correlation coefficients and delta values > 0.70. In a second phase, the questionnaire demonstrated test–retest reliability greater than 60% [43].

The questionnaire evaluates several aspects, including sociodemographic data, skin color and phototype, sun-exposure habits, and history of sunburns. To determine Fitzpatrick skin types, participants were asked about their sensitivity to sunburn and ease of tanning. Sun-protection measures (practices) were assessed using a Likert-type frequency scale (6 items); attitudes toward sun exposure were assessed with a Likert-type agreement scale (10 items); and knowledge about the sun and skin cancer was evaluated with true/false responses (10 items).

A semi-structured interview guide was used, focusing on daily activities, risk, understanding of sun-protection knowledge, understanding of sun-protection practices, perceived disadvantages of sun-protection behaviors, awareness of health problems related to sun exposure, photosensitivity, information sources, and coping with risks associated with prolonged sun exposure. The interview guide was pilot-tested before actual data collection with patients and members of the public attending the hospital.

Operational definitions of variables

The primary variable of interest was the level of sun-protection knowledge, defined as the total score obtained by participants on the 10 knowledge items of the instrument. Results were classified according to the total score distribution using percentiles. An acceptable level of knowledge was defined as the 60th percentile or higher.

The second variable of interest was sun-protection practices, operationally defined as the percentage of total scores obtained from the 6 items included in the questionnaire. Each item assessed the frequency of preventive measures against sun exposure (e.g., wearing hats or caps, long-sleeved clothing, sunglasses, sunscreen use, and avoidance of midday exposure). Practices were classified as inadequate (scores 49.9% or lower) and adequate (scores 50% or higher), cut-off points used in previous studies [44, 45].

Covariates of interest included age, gender, marital status, education level, place of residence, years of work experience, type of agrarian health insurance affiliation, and skin phenotype.

Data analysis

The database created in Microsoft Excel 2010® was imported into Stata/SE® version 17.0 (StataCorp, TX, USA) for data analysis. Qualitative variables were presented using absolute and relative frequencies, while quantitative variables were summarized using measures of central tendency and dispersion.

A bivariate analysis was conducted to evaluate factors associated with an adequate level of knowledge using chi-square tests for categorical variables and Student's t-tests for numerical variables. Strength of association was assessed using Poisson regression with robust variance to estimate prevalence ratios (PR) with 95% confidence intervals (95% CI). A p-value < 0.05 was considered statistically significant.

Qualitative data were analyzed using thematic analysis. Interview transcripts were entered into an Excel matrix that included the main thematic axes from the semi-structured interview guide. The matrix was expanded as new themes emerged from the interviews, generating new columns with additional thematic categories. Each interview lasted between 45 and 60 min. ATLAS.ti® software was used.

Methodological integration was achieved through triangulation of quantitative and qualitative findings, examining areas of convergence, complementarity, dissonance, or silence, according to established criteria [46]. To ensure appropriate integration between phases, the inclusion and exclusion criteria for the quantitative sample were identical to those used for the qualitative sample.

Results

A total of 203 farmers were surveyed throughout the study; all selected participants completed the survey, and no losses occurred during sampling.

The population had a mean age of 59 years, the majority were women (60.1%), and most participants were married or living with a partner (79.3%). The median number of years working in agriculture was 17, and most respondents were from Bagua Grande (70.4%).

Regarding natural skin color, most participants reported having brown skin (75.0%) and predominantly Fitzpatrick phototype IV (35.6%). Additionally, 60.9% reported sun exposure of 3 to 4 h per day.

With respect to sun-protection practices, always or frequently wearing long-sleeved shirts and long pants was common (42.4%); however, avoiding midday sun exposure was uncommon (2.5%). Only 9.8% of participants frequently used sunscreen.

An acceptable level of knowledge was observed in 17.2% of respondents. More details are presented in Table 1.

Regarding the ten knowledge questions, we observed that the vast majority (93.1%) believed that staying in the shade eliminates the risk of radiation-related effects, and 84.6% were unaware that sun-protection measures should be used when the UVI is higher than 3. Participants did

Table 1 Descriptive analysis of the characteristics of the study sample ($n = 203$)

Features	*N* (%)
Age (years)*	59 ± 15.2
Gender	
Male	122 (60.1)
Female	81 (39.9)
Marital status	
Single	18 (8.9)
Married/Cohabitating	161 (79.3)
Widoweb/Divorced	24 (11.8)
Educational level	
None	1 (0.5)
Primary	73 (36.0)
Secondary	105 (51.7)
Superior	24 (11.8)
Length of service (years)**	17.0 (10–24)
Source	
Bagua Grande	143 (70.4)
Morropón-Winguío	1 (0.5)
Santa Elena	4 (2.0)
Huarango	2 (1.0)
La Victoria	17 (8.4)
La Versalla	23 (11.2)
Jahuanga	4 (2.0)
San Antonio	2 (1.0)
Collicate	3 (1.5)
San Miguel	3 (1.5)
Los Pukios	1 (0.5)
Natural skin color	
White	1 (0.5)
Light	40 (19.7)
Tanned	154 (75.9)
Dark	8 (3.9)
Skin reaction when exposed to midday sun for one hour in earl summer	
I never burn, and I tan intensely after 1 week	2 (1.1)
I rarely burn, and I tan very easily	32 (16.8)
I rarely burn, I tan easily	55 (28.8)
Sometimes I burn, I tan gradually	68 (35.6)
Normally burns, tans with difficulty	26 (13.6)
Always burns, never tans	8 (4.1)
Hours per day of sun exposure	
0–2 h	52 (25.7)
3–4 h	123 (60.9)
5 h or more	27 (13.4)
Shade/umbrella use	
Always	0 (0.0)
Usually	9 (4.5)
Sometimes	11 (5.5)
Almost never	14 (7.0)
Never	166 (83.0)
I wear sunglasses	
Always	0 (0.0)
Usually	3 (1.5)
Sometimes	11 (5.4)
Almost never	11 (5.4)
Never	178 (87.7)
I wear a hat or cap	
Always	9 (4.5)
Usually	83 (41.1)
Sometimes	110 (54.4)
Almost never	0 (0.0)
Never	0 (0.0)
I wear a T-shirt with sleeves and long pants	
Always	7 (3.5)
Usually	79 (38.9)
Sometimes	112 (55.1)
Almost never	4 (2.0)
Never	1 (0.5)
I avoid midday hours (12:00 to 16:00 h)	
Always	0 (0.0)
Usually	5 (2.5)
Sometimes	67 (33.5)
Almost never	124 (62.0)
Never	4 (2.0)
Use of sunscreen creams	
Always	1 (0.5)
Usually	19 (9.3)
Sometimes	75 (36.9)
Almost never	60 (29.6)
Never	48 (23.7)
Knowledge level	
Acceptable	35 (17.2)
Under	168 (82.8)

† Some variables may sum to less than 203 due to missing data

*Mean ± Standard deviation

**Median (Interquartile range)

know that infants under one year of age should not be directly exposed to the sun (83.2%) and that dark clothing provides more protection from the sun than light-colored clothing (73.9%). More details are presented in Table 2.

In the bivariate analysis, we observed that older participants had lower levels of knowledge (48 vs. 58 years; $p < 0.001$). Regarding gender, although the difference was not statistically significant ($p = 0.060$), a higher proportion of women demonstrated an acceptable level of knowledge compared to men (74.3% vs. 25.7%). Work experience also had an influence, with an acceptable level of knowledge observed among those who had more years working in agriculture ($p = 0.015$). This may suggest that work experience could be related to greater awareness of sun protection.

The use of sunscreen showed a significant association ($p = 0.010$); most individuals with low levels of knowledge reported lower sunscreen use compared with those who

Table 2 Items assessing the level of knowledge about sun protection

#	Premise (Correct answer)	Correct	Incorrect
1	The use of UVA cabins before the age of 30 increases the risk of melanoma (T)	112 (55.5)	90 (44.5)
2	Ultraviolet radiation causes accelerated skin aging and various forms of skin cancer (T)	91 (44.8)	112 (55.2)
3	In the shade, we are not at risk from the effects of solar radiation (F).	14 (6.9)	188 (93.1)
4	Using sunscreens is the best way to protect yourself from the sun and prevent skin cancer (F)	37 (18.5)	163 (81.5)
5	Once the skinis tanned it is not necessary to use sunscreen (F).	61 (30.5)	139 (69.5)
6	Babies less than 1 year old should not be exposed to direct sunlight (T)	168 (83.2)	34 (16.8)
7	It is necessary to use extreme sun protection measures when the UVI is higher than 3 (T).	31 (15.4)	171 (84.6)
8	Dark clothing protects from the sun more than light clothing (T)	150 (73.9)	53 (26.1)
9	It is advisable to get at least one hour of sunshine a day to ensure adequate vitamin D levels (F).	122 (60.7)	79 (39.3)
10	Children should use sunscreens with a sun protection index equal to or higher than 30 (T).	76 (37.6)	126 (62.4)

V True, *F* False

always or frequently applied it (92.9% vs. 7.1%). More details are presented in Table 3.

First, age showed a significant inverse association with acceptable knowledge (PR: 0.96; 95% CI: 0.95–0.98; $p < 0.001$), indicating that as age increases, the probability of having adequate knowledge about sun protection decreases by 4%. Work experience did not show a significant association with acceptable knowledge ($p = 0.988$).

Regarding natural skin color, individuals with darker tones (Brown–Black) were less likely to have acceptable knowledge compared to those with lighter tones (White–Light) (PR: 0.30; 95% CI: 0.17–0.53; $p < 0.001$). Sun exposure was not significantly associated with acceptable knowledge. Infrequent use of sunscreen was also significantly associated with lower likelihood of having acceptable knowledge, compared with habitual use (PR: 0.37; 95% CI: 0.19–0.70; $p = 0.002$). More details are presented in Table 4.

Below, we present the main findings of the qualitative study. It is important to recall that coded identifiers have been assigned to participants' verbatim quotes to respect their confidentiality.

Daily activities

The interviewees, both men and women, engage in agricultural activities either for self-consumption or for sale, averaging six to eight hours of work per day. In the case of women, these activities are combined with domestic and caregiving responsibilities.

> *"I leave food at the field, check on the animals, wake up at 4 am, wash, cook. Until 10. In the field I plant green onions, beans, cilantro, rice, bananas, lemons. On Thursdays and Fridays I go sell chicken broth and eggs." (P1, woman, Los Puquios).*
> *"In the field I plant corn, I work for myself, for self-consumption. Sometimes two hours, until I have to cook." (P2, woman, El Ron).*

Risk

Despite spending many hours exposed to the sun, participants mentioned various risks associated with their work, although only two specifically stated that prolonged sun exposure is a risk factor.

> *"...the sun burns strongly and I've already been 30 years in agriculture." (P3, man, Quebrada Seca).*
> *"...the sun burns too much. I can't work comfortably. I wear a hat, shorts, a sweatshirt (...) but it's annoying because it burns too much." (P4, man, El Zapote).*

Other risks mentioned focused on physical activities such as cuts, the use of machetes, etc. The risk posed by the isolation of the villages was also noted as a dangerous aspect.

Understanding of knowledge about sun protection and information

Most participants reported not being well informed about the importance of sun protection. However, their main means of protection includes long-sleeved shirts, pants, and hats; some people use sunglasses, and almost no one uses sunscreen.

Two participants mentioned that the main thing they had heard about the topic was that without sun protection, there is a possibility of developing skin cancer.

> *"I wear a hat, shorts, a sweatshirt, no creams, no sunscreen. But I don't know more, I haven't informed myself." (P4, man, El Zapote).*
> *"I've heard it causes cancer." "They hardly explain anything at Essalud, they prescribe and don't give information at Essalud. I don't trust them." (P1, woman, Los Puquios).*
> *"They haven't informed us, only my children. They tell me that it can cause skin cancer." (P5, woman, La Victoria).*
> *"They haven't informed me much." (P6, man, Simón Bolívar).*
> *"People don't protect themselves from the sun. I imagine with sunscreen, but I don't use it because it's*

Table 3 Bivariate analysis of population characteristics and level of knowledge

Features	Acceptable	Under	*p*-value
Age(years)*	48 (30–59)	58 (48–67)	< 0.001
Gender**			0.060
Female	26 (74.3)	96 (57.1)	
Male	9 (25.7)	72 (42.9)	
Marital status***			< 0.001
Single	11 (31.4)	7 (4.2)	
Married/Cohabitating	20 (57.1)	141 (83.9)	
Widowed/Divorced	4 (11.4)	20 (11.9)	
Educational level***			< 0.001
None	0 (0.0)	1 (0.9)	
Primary	63 (71.6)	10 (8.7)	
Secondary	13 (14.8)	92 (80.0)	
Higher	12 (13.6)	12 (10.4)	
Length of service*	15 (10–24.5)	10 (5–22)	0.015
Natural skin color**			< 0.001
White-Light	16 (45.7)	25 (14.9)	
Tanned-Dark	19 (54.3)	143 (85.1)	
Skin reaction when exposed to midday sun for one hour in early summer***			0.87
I never burn, and I tan intensely after 1 week	0 (0.0)	2 (1.3)	
I rarely burn, and I tan very easily.	7 (21.9)	25 (15.7)	
I rarely burn, I tan easily.	11 (34.4)	44 (27.7)	
Sometimes I burn, I tan gradually.	10 (31.3)	58 (36.5)	
Normally burns, tans with difficulty	3 (9.4)	23 (14.5)	
Always burns, never tans	1 (3.1)	7 (4.4)	
Hours per day of sun exposure***			0.443
0–2 h	8 (22.9)	44 (26.3)	
3–4 h	20 (57.1)	103 (61.7)	
5 h to more	7 (20.0)	20 (12.0)	
Shade/Use of umbrella****			0.194
Always/Habitually	3 (8.6)	6 (3.6)	
Infrequent	32 (91.4)	159 (96.4)	
I wear sunglasses***			0.435
Always/Habitually	1 (2.9)	2 (1.2)	
Infrequent	34 (97.1)	166 (98.8)	
I wear a hat or cap**			0.693
Always/Habitually	17 (48.6)	75 (44.9)	
Infrequent	18 (51.)	92 (55.1)	
I wear a T-shirt with sleeves and long pants**			0.233
Always/Habitually	18 (51.4)	68 (40.5)	
Infrequent	17 (48.6)	100 (59.5)	
I avoid midday hours (12:00 to 16:00 h)***			0.610
Always/Habitually	1 (2.9)	4 (2.4)	
Infrequent	33 (97.1)	162 (97.6)	
Use of sunscreen creams			0.010
Always/Habitually	8 (22.9)	12 (7.1)	
Infrequent	27(77.1)	156 (92.9)	

Median (interquartile range); *Mann-Whitney test; **Chi-square test; ***Fisher's exact test; *Maximum (interquartile range); *Mann-Whitney test; **Chi-square test; ***Fisher's exact test

expensive, 60 soles. Long sleeves, hat I imagine, but they haven't informed me." (P2, woman, El Ron).

Understanding of sun protection practices

All participants practice some form of sun protection such as wearing long-sleeved shirts, pants, and hats; some also use sunglasses. Sunscreen use is infrequent, mainly due to its high cost for this population. Only one participant mentioned using it.

"I take care of myself with long clothing, shirt, pants, sneakers, hat, socks." (P7, woman, Higos Urco).
"I use a hat, long-sleeved shirt, pants, we don't use sunscreen, it drips off with sweat." (P3, man, Quebrada Seca).
"People don't protect themselves from the sun. I imagine with sunscreen but I don't use it because it's expensive, 60 soles." (P2, woman, El Ron).
"Hat, shirt and long pants. I don't use sunscreen, it's expensive." (P6, man, Simón Bolívar).

Understanding of barriers to sun protection

Although participants reported using appropriate clothing to protect themselves from sun exposure, they also commented that wearing so much clothing can be uncomfortable and time-consuming, especially for younger individuals, causing annoyance. In addition, sunscreen remains difficult to use due to its high cost. Most participants also expressed a lack of knowledge about key sun protection measures or their importance.

"I don't use sunscreen because I can't afford it. I have to invest in products for the plants. I live isolated in a village." (P7, woman, Higos Urco).
"Many people are bothered by wearing so much clothing; buying creams is expensive, 40 or 50 soles." (P3, man, Quebrada Seca).
"I take care of myself because my children worry about me, like using sunglasses, hats, long sleeves, but sometimes because of time my son doesn't pay attention and doesn't take care of himself." (P5, woman, La Victoria).

Knowledge about health problems related to sun exposure

Of all participants, only one person reported not having heard of or experienced health problems related to sun exposure.

"Tiredness, fatigue, sometimes there's no water and you get desperate, you go to the shade because you can't stand the sun. When the sun shines on my face, it feels like it burns my skin." (P7, woman, Higos Urco).
"Skin issues, spots appear, I'm getting little spots on my face but I haven't had it checked. I prefer home remedies, they're more accessible." (P4, man, El Zapote).

Health problems

All participants reported health problems related to sun exposure and also to the lack of access to clean drinking water.

"Skin issues, spots appear, I'm getting little spots on my face but I haven't had it checked. I prefer home remedies, they're more accessible." (P4, man, El Zapote).
"I have a spot on my face, but my spots are from the sun and from water from streams, we don't have drinking water." (P3, man, Quebrada Seca).
"My children get rashes from the heat, me too, my body itches." (P1, woman, Los Puquios).
"My son doesn't take care of himself, his skin gets red, it turns dark. My husband is burned from the sun, his skin peels." (P5, woman, La Victoria).
"Sometimes I get allergies, itching and burning on my arms, back and chest." (P6, man, Simón Bolívar).

Photosensitivity

None of the participants knew whether they were photosensitive, and some had incorrect notions about the concept, as in the case of one woman:

"What is that? I don't know, is it the eyesight? I wear glasses, the doctor told me I had dry eye, she prescribed glasses about 10 years ago; they darken when I go into the sun, but indoors they lighten. I can't go out in the sun without glasses." (P5, woman, La Victoria).

Coping, care, and access to health services

Participants reported that when facing a skin problem, they usually apply creams or take pills. However, they expressed overall dissatisfaction with their local health facilities and hospitals in the event of a health emergency. Access is one of the main barriers, as the distance makes frequent visits difficult. The treatment provided by health workers and long waiting times were also key barriers. For this reason, some participants prefer to self-medicate.

"I have to go down to the town by transport, an hour and a half. You must get an appointment a day in

Table 4 Association between population characteristics and acceptable level of knowledge about sun protection

Characteristics		PR	95% IC	*p*-value
Age(years)*		0.96	0.95–0.98	< 0.001
Gender				
	Female	Ref		
	Male	0.52	0.26–1.01	0.070
Marital status				
	Single	Ref		
	Married/Cohabitating	0.20	0.12–0.35	< 0.001
	Widowed/Divorced	0.27	0.10–0.72	0.009
Educational level				
	Primary	Ref		
	Secondary	0.90	0.42–1.95	0.797
	Superior	3.65	1.81–7.37	< 0.001
Length of service**				
	0 a 20	Ref		
	21 a 45	1.01	0.50–2.01	0.988
Natural skin color				
	White-Light	Ref		
	Tanned-Dark	0.30	0.17–0.53	< 0.001
Skin reaction when exposed to midday sun for one hour in early summer				
	I never burn, and I tan intensely after 1 week	Ref		
	I rarely burn, and I tan very easily.	0.91	0.39–2.13	0.835
	I rarely burn, I tan easily.	0.67	0.028–1.61	0.372
	Sometimes I burn, I tan gradually.	0.53	0.15–1.85	0.317
	Normally burns, tans with difficulty	0.57	0.81–4.02	0.574
Hours per day of sun exposure				
	0–2 h	Ref		
	3–4 h	1.06	0.5–2.25	0.886
	5 h to more	1.69	0.68–4.16	0.258
Shade/Use of umbrella				
	Always/Habitually	Ref		
	Infrequent	0.50	0.19–1.34	0.168
I wear sunglasses				
	Always/Habitually	Ref		
	Infrequent	0.51	0.10–2.61	0.419
I wear a hat or cap				
	Always/Habitually	Ref		

Table 4 (continued)

Characteristics		PR	95% IC	*p*-value
	Infrequent	0.89	0.48–1.62	0.693
I wear a T-shirt with sleeves and long pants				
	Always/Habitually	Ref		
	Infrequent	0.69	0.38–1.27	0.236
I avoid midday hours (12:00 a 16:00 h)				
	Always/Habitually	Ref		
	Infrequent	0.85	0.14–5.04	0.854
Use of suncreen creams				
	Always/Habitually	Ref		
	Infrequent	0.37	0.19–0.70	0.002

advance. They prescribed a cream and it hasn't gone away. There's no drinking water, my fungal infection came back. They told me to return in a month. I went to the Samaritan clinic in Bagua Grande. In the countryside there are always animals that bite and there are no health centers nearby. From the sun, from the heat sometimes, but just a little. Fungal infection on my eyelid, on both eyes. The cream relieves the itching; without it nothing helps, if I scratch it gets wounded. I prepare my matico water myself." (P7, woman, Higos Urco).

"I looked for a virtual dermatologist in Lima, a teleconsultation; I have a spot on my face and he told me to stop working for a few days, but how am I not going to work? He tells me not to work, but I can't. I had the cream they prescribed made in Chiclayo, the travel was expensive and it didn't work." (P3, man, Quebrada Seca).

"To go to the health center you have to be dying, Bagua Grande, I went once, my son got ill, you have to be dying, better to go to a pharmacy, the hospital is useless." (P2, woman, El Ron).

"...medication, small pills. They hardly explain anything at Essalud, they prescribe and don't give information at Essalud. I don't trust them. Only some of them; the latest doctors don't explain anything. They referred my husband to Chiclayo because his ulcer burst, but he went to Chiclayo and they didn't do anything and scheduled an endoscopy, and it was an emergency, he was bleeding, but they did nothing. It's been more than a year since his ulcer burst and they still haven't done the endoscopy." (P1, woman, Los Puquios).

"Pills at the hospital, sometimes they help, but they don't relieve me completely." (P6, man, Simón Bolívar).

Discussion

Farmers in Bagua Grande are chronically exposed to sunlight because the town is located in the tropics, where UV intensity remains high throughout the year, and because of the work they do. In addition, sun protection programs lack widespread institutional adoption in current Peruvian public health policies, and little work is being done in the field of primary health care, especially in the prevention of skin cancer. Instead, advertising and publicity by cosmetics companies have led to people having inaccurate knowledge about sun protection [47–49].

In this study, farmers had a low level of knowledge about photoprotection. These findings are consistent with the results of studies by Zink et al. [24] in Germany and Hogan et al. [25] in Ireland, where low levels of knowledge were also reported, despite the fact that these countries have relatively lower environmental levels of ultraviolet radiation. However, our results differ from the findings of the study by D'Souza et al. [50] on rural farmers in Australia, who showed a high level of knowledge about sun safety. This discrepancy is not explained by climatic differences, as Australia has high levels of UV radiation, similar to or higher than those in South America, but rather by sociocultural and contextual factors. In Australia, there is greater public awareness of skin cancer, sustained educational campaigns, and labor regulations that promote sun protection in agricultural activities. In contrast, in the Peruvian rural context, limited access to preventive information, reduced risk perception, and differences in the predominant skin type could influence the low level of knowledge observed. This reinforces the need for research that delves deeper into the sociocultural factors associated with sun exposure and photoprotection practices in Latin American rural environments.

Although a few studies conducted in the national context show partly similar [51] or opposite [29] results. However, we can affirm that these studies used instruments that could have induced measurement biases as they had not been fully validated. In the case of our study, a valid, reliable instrument with good sensitivity to change for the Spanish-speaking population was used [42, 52]. In this regard, more research should be conducted in the national context on sun exposure in farmers using better methods for a better conceptualization of the results.

The age distribution in the sample showed that farmers with an acceptable level of knowledge had a median age of 48 (IQR: 30–59), while those with low knowledge had a median age of 58 (IQR: 48–67), a difference that was statistically significant ($p < 0.001$). This result suggests that younger farmers tend to have a higher level of knowledge about sun exposure and photoprotection measures. However, given that most participants were middle-aged, the findings may mainly reflect the behaviors and

perceptions of this age group. Consequently, the generalization of the results to younger farmers should be done with caution. It is recommended that future studies include a larger number of young participants or analyze generational differences in sun safety practices in a stratified manner.

The literature shows mixed results regarding the relationship between age and knowledge about photoprotection, with some studies showing a knowledge deficit in younger adults [24] and others in older adults [53]. In our context, given that this is an age group between 48 and 67 years old with a low level of knowledge (58%), it is likely that the limited information they possess does not allow them to engage in effective sun protection practices, especially if the information was not provided with adequate strategies or is not relevant to the individual because it does not respond to their interests. This result is consistent with the knowledge-related barriers identified by other authors [54].

We can conjecture that age alone does not seem to determine preventive behavior against sun exposure; that is, even when there is awareness of the risks, the adoption of protective measures may be conditioned by practical and economic barriers. As the interviewees mentioned in relation to their work, it is tedious to take certain measures to protect themselves from the sun because too much clothing is overwhelming and hinders their work, and the purchase of sunscreen is beyond their budget. Despite the presumed damage from the sun, in some cases with more than 30 years in agriculture, people do not act consciously and many do not follow this advice. These findings show that material limitations and working conditions have a decisive influence on the practice of photoprotective measures, beyond chronological age.

In the qualitative findings, participants also mentioned skin problems such as spots, irritations, or fungal infections, as well as a lack of access to drinking water. Although these conditions are not directly related to ultraviolet radiation, they reflect the environmental and sanitary conditions in which agricultural work is carried out. The combination of prolonged sun exposure, high temperatures, humidity, and limited hygiene conditions favors the appearance or worsening of dermatological conditions, which can sometimes be confused with the effects of UV radiation. This overlap of factors contributes to a vague perception of solar risk among farmers, who tend to attribute skin lesions to "sun," "water," or "heat" without clearly distinguishing their origin. Understanding these local interpretations is essential for the design of educational and preventive interventions, as the perception of skin damage conditions sun protection behaviors and the use of appropriate preventive measures.

In relation to natural skin color and level of knowledge, individuals with darker skin tones were less likely to achieve acceptable levels of knowledge about photoprotection. A study conducted in black Hispanic and non-Hispanic populations showed that the majority perceived a low risk of skin cancer due to their darker skin tone [55]. These results differ from the findings of Agarwal et al. [10] in India, where participants demonstrated greater knowledge about the effects of sun exposure. However, it is important to consider that this study was conducted in a different sociocultural and economic context, where intense solar radiation coexists with greater public awareness and access to health campaigns.

In our rural context, dark-skinned farmers tend to underestimate their vulnerability, as health education campaigns have historically been targeted at fair-skinned populations [56, 57]. Likewise, the interviewees' narratives reveal that the limited dissemination of information on the harmful effects of UV radiation by health centers leads to a lack of awareness about the need for photoprotection according to skin type, increasing the future risk of skin cancer. In this regard, public health and educati campaigns have failed to adequately address the specific needs of farmers [21].

Regarding skin type classification, our study used the Fitzpatrick skin type scale, widely used in dermatological practice and epidemiological research on photoprotection. While this system is a practical tool based on skin color and response to the sun in terms of burning and tanning, it has limitations when applied to populations with darker skin, where self-perception of skin tone may differ from objective measurements [58, 59]. In addition, studies have shown that people tend to overestimate the darkness of their skin or underestimate its erythemal sensitivity [60]. Despite these limitations, the Fitzpatrick system remains the most widely used in clinical settings due to the absence of an alternative system that is universally validated and culturally adaptable for all ethnic groups [58, 61]. In this study, the burn and tan categories were maintained in accordance with the CHACES instrument, with the terms "tan" being explained to participants as darkening of the skin after sun exposure and "burn" as redness or skin damage associated with UV [62, 63], in order to ensure understanding of the questions. This strategy reduced potential interpretation biases and preserved the validity and reliability of the instrument used. Therefore, health promotion strategies must integrate culturally sensitive components adapted to the economic conditions of agricultural communities, so that photoprotection is perceived as a relevant and achievable practice.

Consequently, scenarios similar to this population require educational interventions focused on specific stages of life. It is necessary to specify that interventions

in photoprotection should not be generic; they should be targeted at certain characteristics of the population such as gender, age, skin phototype, level of education, type of agriculture practiced, and dermatological needs. The latter should take into account the population under study, given that mature skin has different needs and may suffer cumulative damage due to prolonged exposure, making it more susceptible to burns.

In this study, more than half of the farmers were exposed to the sun for 3 to 4 h. This differs from other studies that report that the average working day was 7 h [64] or 9.4 h [22]. At the same time, most agricultural workers are exempt from policies regulating maximum hours per shift and the application of health and safety measures, as farmers are self-employed and therefore these regulations do not apply [24]. One study found that older farmers worked long hours on agricultural tasks [65]. Another study found that women living in rural areas may work long hours, as they may help on the farm, as well as perform domestic tasks and raise children [66]. Although there was no statistically significant association between hours of exposure and knowledge, we did observe that people who worked longer hours per day had less knowledge about sun protection. However, this contradicts the findings of other studies [67]. This finding indicates that specific groups or climatic variations in geographical areas may influence this association.

We observed that the majority of participants (93.1%) believed that being in the shade meant they were not at risk of suffering the effects of solar radiation. This perception is not entirely correct, since although shade significantly reduces direct exposure to UV rays, radiation reflected by surfaces such as the ground, sand, or water can still reach the skin [6]. Furthermore, in natural shade such as that provided by trees, UV rays can filter through the leaves, reducing the effectiveness of the protection [5]. However, shade should not be considered an ineffective measure. On the contrary, it represents a simple, free, and sustainable form of photoprotection, especially useful for farmers who work outdoors for long periods. In certain circumstances, shade can offer more consistent protection than sunscreen, whose effect depends on proper application and reapplication, or than protective clothing, which can be uncomfortable in hot climates. Ideally, workers should avoid direct exposure during the hours of peak solar radiation; however, this is not always feasible due to work demands. Therefore, it is recommended to promote the combined use of measures seeking or creating shade, wearing wide-brimmed hats, long-sleeved clothing, and sunscreen as a comprehensive photoprotection strategy adapted to the actual conditions of agricultural work.

Another relevant aspect to consider is the influence of albedo, understood as the ability of surfaces to reflect solar ultraviolet radiation [68]. Agricultural surfaces can reflect different levels of UV radiation depending on their composition, color, moisture, and vegetation cover [69], the type of crop, the varieties used, and agricultural practices [70]. In the context of this study, farmers worked mainly in rice fields, where moist soil or low vegetation cover generates intermediate levels of reflectance [71], thus affecting the cumulative exposure of workers. This phenomenon can increase the total dose of radiation received, even when workers seek shade or wear protective clothing. Therefore, sun safety strategies aimed at this occupational group should consider not only direct exposure but also reflected radiation, promoting complementary measures such as the use of wide-brimmed hats, face shields, and frequent reapplication of sunscreen.

In the present study, 84.6% of respondents were unaware that they should use sun protection measures when the UV index is above 3. This is corroborated by other studies that reported similar results [72]. Farmers need to be provided with information about the UV index to improve sun protection [32] and taught how to check the daily UV index using the weather app on their phone [22].

It was observed that 42.4% of respondents practiced photoprotection by wearing long-sleeved shirts and long pants. This coincides with other studies that have reported that the most common sun protection practice was the use of long pants and long-sleeved shirts [47, 73]. However, in this regard, it is necessary to take into account the opinions of some farmers who mention discomfort when wearing long pants and shirts or feeling too hot when wearing sun protection clothing, which is related to the studies by El-Shafei & Said [22], Trenerry et al. [54], and Carley & Stratman [72], respectively, providing an explanation as to why the vast majority do not use them.

Another worrying fact is the high percentage of people who are exposed to the sun during peak radiation hours (64%). A systematic review found that it was impractical for farmers to avoid the sun during peak hours [21]. It therefore seems reasonable to prioritize adaptation measures in the workplace, such as appropriate heat breaks, shade, and hydration, to safeguard the health and well-being of agricultural workers [74].

According to the survey results, 87.7% of participants did not wear sunglasses while working in intense sunlight. This finding is inconsistent with the results of the study by D'Souza et al. [50]. The interviews also showed that the attitude toward eye protection is often ignored and specific guidance on sun exposure is not emphasized in farmers' awareness, and attention would be paid to protecting themselves from dust or dirt and dry eye disease. Rural residents are exposed to higher levels of UV than urban residents due to less shade; excessive

exposure is directly associated with eye diseases [75]. This result probably responds to the dissemination of information with little connection to and emphasis on the incidence of eye disease risk from excessive UV exposure, which, on the contrary, tends to focus on skin damage. In fact, the participants in this study rarely associated it with macular damage, which was reason enough not to be persuaded to change their habits.

In particular, the use of sunscreen was underestimated, with price being the main reason for not applying it. Only 9.8% of participants frequently used sunscreen. Studies have found that daily sunscreen use is below 20% in Australian adults [33]. One study found that the prevalence of sunscreen use was low at 21.4% among farmers in the United States [9]. In a study conducted in Iran by Panahi & Salehi [64], most rice farmers did not use sunscreen. According to a study by Cueva-Puelles et al. [29], 63.0% of Peruvian farmers did not use sunscreen as a sun protection measure. Infrequent use of sunscreen was associated with less acceptable knowledge. Therefore, it is very important to raise awareness about photoprotection methods and their proper application among farmers, given their frequent overexposure to UV radiation [73]. A more in-depth exploration of discomfort revealed that the texture of sunscreen with dust and dirt that accompanies agricultural work is undesirable [50]. On the other hand, there are other factors, such as cost and the greasiness of the product, that prevent optimal use of sunscreen [10]. Today, sunscreens are available in less greasy formulations, such as gels, sprays, and lip balms, but people are unaware of this [10]. Alcohol-based sunscreen sprays, gels, and lotions are not as heavy or greasy as creams and are better tolerated by people [73]. In addition, awareness should be raised about the essential reapplication of sunscreen after 30 min, especially for those who work outdoors [73].

Participants did know that babies under one year of age should not be exposed to direct sunlight (83.2%) and that dark clothing protects from the sun more than light clothing (73.9%). The American Academy of Pediatrics has stated that babies under 6 months of age should be protected from the sun and that vitamin D requirements should be met through diet and fortified foods rather than deliberate exposure to the sun [6]. Dark colors absorb more ultraviolet radiation and therefore provide greater protection than light colors [76].

In relation to gender, although the difference was not statistically significant (p = 0.060), a higher percentage of women were found to have acceptable sun protection compared to men (74.3% vs. 25.7%). Similarly, D'Souza et al. [50] reported that women were more likely to respond correctly than men (p = 0.032). Seité et al. [77] also found that women appeared to be better informed than men (p < 0.001). Although it remains to be evaluated, it can be assumed that women pay more attention to their skin and appearance, making them more likely to protect themselves from the sun. Work seniority does not have a significant effect on knowledge in the univariate analysis, but it is significant in the group with acceptable knowledge in the bivari t analysis. This could indicate that work experience may be related to greater awareness of sun protection.

As in previous studies in Peru [33, 76] and Ecuador [78], the "CHACES" questionnaire developed in Spain was administered in Spanish, given the lack of a validated questionnaire for the study population. Although in those studies the questionnaire was administered to Peruvian beachgoers and Ecuadorian teachers and university students, rather than farmers, this instrument was chosen because of its cultural adaptation to the Latin American context and its relevance for assessing knowledge about sun exposure. The inclusion of the question on the use of tanning beds, although these are not common in the rural population studied, but are in some urban areas of the country, was based on the need to assess knowledge about the carcinogenic risks of UVA rays, to which farmers are exposed daily through sunlight. It is important to note that during the administration of the questionnaire, additional clarifications were provided to ensure understanding of the use of artificial tanning equipment. This allows us to affirm that the inclusion of this question contributes to a broader understanding of knowledge about sun exposure in the study population, without compromising the validity of the results obtained.

A major limitation of the present study is that the questions related to sun protection practices were general in nature, given that items from the validated "CHACES" questionnaire were used, which assesses the frequency of use of hats, sunglasses, longsleeved clothing, and sunscreen without specifying the technical characteristics of these items. Therefore, it was not considered whether the sunglasses had an ultraviolet radiation filter, whether the hats were wide-brimmed or cap-type, or whether the garments had a fabric dense enough to block UV radiation. Nor was the sun protection factor (SPF), the frequency of reapplication, or the amount of sunscreen used inquired. This lack of detail limits the interpretation of the actual effectiveness of the reported measures. It is recommended that future research complement validated instruments with direct observations or more specific questions about the characteristics of protective elements in order to obtain a more accurate assessment of sun safety practices in agricultural populations.

Furthermore, this study may be subject to certain biases. Selection bias could have occurred due to the voluntary participation of farmers; however, this risk was reduced in the quantitative phase, as a simple random sample of 203 agricultural workers was used.

In the qualitative phase, participants were selected from among the same farmers who participated in the quantitative survey, which helped to maintain the representativeness of the group and reduce selection bias. Another selection bias stems from the fact that only farmers with health insurance who used hospital services were included. However, the study results may be extrapolated to agricultural insurance holders in climatic and geographical conditions similar to those in Bagua Grande, meaning cities located in jungle regions with a small number of inhabitants. Information bias may also have influenced the results, as self-reported data depend on the accuracy of participants' memory and understanding. Likewise, social desirability bias can not be ruled out, as some farmers may have overestimated their sun protection practices in order to provide socially acceptable responses. Even so, the results provide relevant information for understanding and improving sun safety practices among agricultural workers exposed to ultraviolet radiation. These limitations should be considered when interpreting the findings and reinforce the need to employ methodological triangulation strategies in future research on sun safety among agricultural workers.

Conclusions

Farmers' knowledge about sun protection was low and their practices were inadequate, while the identified barriers hinder the implementation of sun-safety behaviors. Therefore, strategic interventions aimed at modifying behaviors and photoprotection practices—designed according to the characteristics of the population and delivered through clear instructions adapted to farmers' individual needs and attitudes—are required. Primary and secondary prevention should be strengthened through health promotion campaigns, educational initiatives, and screening programs, among others, to reduce the incidence and promote early diagnosis of skin cancer, particularly targeting rural populations, in which the barriers identified in this study were found.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12889-026-26202-0.

Supplementary Material 1.

Authors' contributions

Nelson Cesar Santisteban Salazar: Conceptualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Supervision. Mirtha Yvis Santisteban Salazar: Formal analysis, Writing – original draft, Writing – review & editing. Magnolia Anacarina Arrasco Barrenechea: Formal analysis, Writing – original draft, Writing – review & editing.

Funding

This study was funded by the Social Health Insurance of Peru (EsSalud) through the Kaelín Award in Health Sciences.

Data availability

The datasets generated and analyzed during this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The study was conducted in accordance with the Declaration of Helsinki. All participants were informed about the purpose of the study and the protection of their privacy, and written informed consent was obtained. The confidentiality of all personal data not disclosed in this document was ensured.
The study protocol was approved by the Research Ethics Committee of the Hospital Nacional Almanzor Aguinaga Asenjo of the Red Prestacional Lambayeque, Seguro Social de Salud del Perú (NIT-7550-2020-052).

Competing interests

The authors declare no competing interests.

Received: 10 April 2025 / Accepted: 5 January 2026
Published online: 16 January 2026

References

1. Modenese A, Korpinen L, Gobba F. Solar radiation exposure and outdoor work: an underestimated occupational risk. Int J Environ Res Public Health. 2018;15(10):2063. https://doi.org/10.3390/ijerph15102063.
2. Leiter U, Keim U, Garbe C. Epidemiology of skin cancer: update 2019. Adv Exp Med Biol. 2020;1268:123–39. https://doi.org/10.1007/978-3-030-46227-7_6.
3. Rodríguez-Zamorano P, Puebla-Tornero L, Martín-Santos L, Román-Villaizán M, Guerra-Tapia A. Estudio transversal sobre la repercusión de la información educativa en los hábitos de fotoprotección tópica en estudiantes de medicina. Med Fam SEMERGEN. 2018;44(6):420–9. https://doi.org/10.1016/j.semerg.2018.02.004.
4. Hübner J, Waldmann A, Geller AC, Weinstock MA, Eisemann N, Noftz M, Bertram S, Nolte S, Volkmer B, Greinert R, Breitbart E, Katalinic A. Interval cancers after skin cancer screening: incidence, tumour characteristics and risk factors for cutaneous melanoma. Br J Cancer. 2017;116(2):253–9. https://doi.org/10.1038/bjc.2016.390.
5. Hung M, Beazer IR, Su S, Bounsanga J, Hon ES, Lipsky MS. An exploration of the use and impact of preventive measures on skin cancer. Healthcare. 2022;10(4):743. https://doi.org/10.3390/healthcare10040743.
6. Jindal AK, Gupta A, Vinay K, Bishnoi A. Sun exposure in children: balancing the benefits and harms. Indian Dermatol Online J. 2020;11(1):94–8. https://doi.org/10.4103/idoj.IDOJ_206_19.
7. Srivastava SB. Vitamin D: do we need more than sunshine? Am J Lifestyle Med. 2021;15(4):397–401. https://doi.org/10.1177/15598276211005689.
8. Ramírez E, Ferrán M, Torres D, Suarez L, Lima R. Protección solar En niños. Rev Med Electron. 2016;38(2):177–84.
9. Ragan KR, Buchanan N, Thomas CC, Tai EW, Sussell A, Holman DM. Skin cancer prevention behaviors among agricultural and construction workers in the United States, 2015. Prev Chronic Dis. 2019;16:E15. https://doi.org/10.5888/pcd16.180446.
10. Agarwal SB, Godse K, Patil S, Nadkarni N. Knowledge and attitude of general population toward effects of sun exposure and use of sunscreens. Indian J Dermatol. 2018;63(4):285–91. https://doi.org/10.4103/ijd.IJD_609_17.
11. Neale RE, Lucas RM, Byrne SN, Hollestein L, Rhodes LE, Yazar S, Young AR, Berwick M, Ireland RA, Olsen CM. The effects of exposure to solar radiation on human health. Photochem Photobiol Sci. 2023;22(5):1011–47. https://doi.org/10.1007/s43630-023-00375-8.
12. Sabando JA, Dos Santos MH, Calvo D. Incidencia Del cáncer de Piel no melanoma En Un Cupo Del Centro de Salud de Zaramaga. Aten Primaria. 2025;47(3):189–90. https://doi.org/10.1016/j.aprim.2014.05.012.
13. Gupta AK, Bharadwaj M, Mehrotra R. Skin Cancer Concerns in People of Color: risk factors and prevention. Asian Pac J Cancer Prev. 2016;17(12):5257–64. https://doi.org/10.22034/APJCP.2016.17.12.5257.
14. Wunderlich K, Suppa M, Gandini S, Lipski J, White JM, Del Marmol V. Risk factors and innovations in risk assessment for melanoma, basal cell carcinoma, and squamous cell carcinoma. Cancers (Basel). 2024;16(5):1016. https://doi.org/10.3390/cancers16051016.

15. Widemar K, Falk M. Sun exposure and protection index (SEPI) and self-estimated sun sensitivity. J Prim Prev. 2018;39(5):437–51. https://doi.org/10.1007/s10935-018-0520-0.
16. Darjania T, Kituashvili T, Galdava G. The impact of education level on basal cell carcinoma development risk. Cureus. 2024;16(6):e61827. https://doi.org/10.7759/cureus.61827.
17. Symanzik C, John SM. Sun protection and occupation: current developments and perspectives for prevention of occupational skin cancer. Front Public Health. 2022;10:1110158. https://doi.org/10.3389/fpubh.2022.1110158.
18. Souto EB, Jäger E, Jäger A, Štěpánek P, Cano A, Viseras C, et al. Lipid nanomaterials for targeted delivery of dermocosmetic ingredients: advances in photoprotection and skin anti-aging. Nanomaterials (Basel). 2022;12(3):377. https://doi.org/10.3390/nano12030377.
19. Salvado M, Fraga A, Marques DL, Pires IM, Gonçalves CC, Silva NM. Sun exposure in pediatric age: perspective of caregiver. Children (Basel). 2021;8(11):1019. https://doi.org/10.3390/children8111019.
20. Castañeda P, Eljure J. El cáncer de piel, Un problema actual. Rev Fac Med UNAM. 2016;59(2):6–14.
21. Smit-Kroner C, Brumby S. Farmers sun exposure, skin protection and public health campaigns: an Australian perspective. Prev Med Rep. 2015;2:602–97. https://doi.org/10.1016/j.pmedr.2015.07.004.
22. El-Shafei DA, Said RM. Sun safety: knowledge and behavior among Egyptian farmers-a multicomponent intervention study. J Cancer Educ. 2023;38(3):1042–9. https://doi.org/10.1007/s13187-022-02230-3.
23. Wright CY, Norval M. Health risks associated with excessive exposure to solar ultraviolet radiation among outdoor workers in South Africa: an overview. Front Public Health. 2021;9:678680. https://doi.org/10.3389/fpubh.2021.678680.
24. Zink A, Schielein M, Wildner M, Rehfuess EA. Try to make good hay in the shade – it won't work!' A qualitative interview study on the perspectives of Bavarian farmers regarding primary prevention of skin cancer. Br J Dermatol. 2019;180(6):1412–9. https://doi.org/10.1111/bjd.17872.
25. Hogan V, Hogan M, Kirwan O, Langan C, McLaughlin C, Moynihan Á, Connolly A, Walsh J, Coggins M. Sun-related knowledge and practices in Irish construction and agricultural workers. Occup Med. 2024;74(5):378–85. https://doi.org/10.1093/occmed/kqae042.
26. Ozkan M, Adana F, Yesilfidan D. Sun protection behaviour of agricultural workers and associated factors in west of Turkey. J Pak Med Assoc. 2022;72(7):1320–4. https://doi.org/10.47391/JPMA.1856.
27. Moeini B, Ezati E, Barati M, Rezapur-Shahkolai F, Mohammad N, Afshari M. Skin cancer preventive behaviors in Iranian farmers: applying protection motivation theory. Workplace Health Saf. 2019;67(5):231–40. https://doi.org/10.1177/2165079918796850.
28. Hault K, Rönsch H, Beissert S, Knuschke P, Bauer A. Knowledge of outdoor workers on the effects of natural UV radiation and methods of protection against exposure. J Eur Acad Dermatol Venereol. 2016;30(Suppl 3):34–7. https://doi.org/10.1111/jdv.13631.
29. Cueva-Puelles LA, Urdanivia-Cotrina MC, Valle-Bedregal A, Aponte-Villacorta JE, Dávila-Requelme LR, Morales-Olivera S, et al. Conocimientos y hábitos sobre protección solar en agricultores del caserío de Macuaco, Chiclayo, Perú, 2019. Rev Med Exp. 2019;5(4):173–8. https://doi.org/10.37065/rem.v5i4.396.
30. Gupta V, Sharma VK. Skin typing: Fitzpatrick grading and others. Clin Dermatol. 2019;37(5):430–6. https://doi.org/10.1016/j.clindermatol.2019.07.010.
31. World Health Organization, Global Solar UV, Index A, Practical Guide. 2002. https://iris.who.int/server/api/core/bitstreams/94a908ec-14ee-4bb4-84eb-ace483c9eb51/content. Accessed 20 May 2024.
32. Heckman CJ, Liang K, Riley M. Awareness, understanding, use, and impact of the UV index: a systematic review of over two decades of international research. Prev Med. 2019;123:71–83. https://doi.org/10.1016/j.ypmed.2019.03.004.
33. Fernandez-Quiroz EL, Gonzales-Chachapoyas L, Alcantara-Diaz AL, Bulnes-Villalta B, Ayala-Porras Z, Toro-Huamanchumo CJ. Sunburns among beachgoers in the northern coast of Peru: frequency and factors associated. PeerJ. 2021;9:e11473. https://doi.org/10.7717/peerj.11473.
34. Delgado G. Radiación solar extrema en Perú: ¿cómo nos afecta? ¿qué distritos son los más dañados y cómo cuidarnos? https://elcomercio.pe/respuestas/como/rayos-uv-radiacion-solar-extrema-en-peru-como-nos-afecta-que-distritos-son-los-mas-danados-y-como-cuidarnos-noticia/?ref=ecr. Consultado el 27 de febrero de 2024.
35. Ministerio de Salud. Boletín Epidemiológico del Perú 2018. https://www.dge.gob.pe/portal/docs/vigilancia/boletines/2018/31.pdf. Consultado el 10 de julio de 2024.
36. Ministerio de Salud. Análisis de la situación del cáncer en el Perú, 2018. https://observateperu.ins.gob.pe/node/1630. Consultado el 11 de setiembre del 2024.
37. Barboza E, Corroto F, Salas R, Gamarra O, Ballarín D, Ollero A. Hidrogeomorfología En áreas tropicales: Aplicación Del índice hidrogeomorfológico (IHG) En El río Utcubamba (Perú). Ecol Apl. 2017;16(1):39–47.
38. American Cancer Society. Mortality rates 2015–2019. https://cancerstatisticscenter.cancer.org/#!/. Accessed 22 September 2024.
39. Irish Cancer Society. One in Four Skin Cancer Deaths Are From the Construction and Farming Sector. https://www.cancer.ie/about-us/news/one-in-four-skin-cancer-deaths-are-from-the-construction-and-farming-sector. Accessed 5 July 2024.
40. World Bank Group. Climate-Smart Agriculture in Peru. https://climateknowledgeportal.worldbank.org/sites/default/files/2019-06/CSA-in-Peru.pdf. Accessed 30 November 2024.
41. Ministerio de la Producción. Diagnóstico Productivo Regional Amazonas. https://www.producempresarial.pe/wp-content/uploads/2024/08/Ficha_Diagnostico_Amazonas_Ene-Abr-2024_05.07.24.pdf Consultado el 30 de Diciembre del 2024.
42. De Troya-Martín M, Blázquez-Sánchez N, Rivas-Ruiz F, Fernández-Canedo I, Rupérez-Sandoval A, Pons-Palliser J, et al. Validation of a Spanish questionnaire to evaluate habits, attitudes, and understanding of exposure to sunlight: "the beach questionnaire." Actas Dermosifiliogr. 2009;100(7):586–95.
43. Blázquez-Sánchez N, Rivas-Ruiz F, Bueno-Fernández S, Arias-Santiago S, Fernández-Morano MT, De Troya-Martín M. Validation of a questionnaire designed to study knowledge, attitudes, and habits related to sun exposure among young adults: the CHACES questionnaire. Actas Dermosifiliogr. 2020;111(7):579–89. https://doi.org/10.1016/j.ad.2020.02.002.
44. Ugwu NI, Oti WJ, Ugwu CN, Uneke CJ. Voluntary non-remunerated blood donation: awareness, perception, and attitude among potential blood donors in Abakaliki, Nigeria. Niger J Clin Pract. 2019;22(11):1509–15. https://doi.org/10.4103/njcp.njcp_159_19.
45. Ren X, Liu Y, Wang X, Li R, Guo X, Zhao S, et al. Gender-, Age-, and Region-Specific associations between obesity and nutrition/health knowledge, dietary diversity, and physical activity in Chinese school-age students: a cross-sectional study. Nutrients. 2025;17(13):2214. https://doi.org/10.3390/nu17132214.
46. O'Cathain A, Murphy E, Nicholl J. Three techniques for integrating data in mixed methods studies. BMJ. 2010;341:c4587. https://doi.org/10.1136/bmj.c4587.
47. Yang EJ, Beck KM, Maarouf M, Shi VY. Truths and myths in sunscreen labeling. J Cosmet Dermatol. 2018;17(6):1288–92. https://doi.org/10.1111/jocd.12743.
48. Naseer M, Asad N, Javaid A, Moughal R, Akram H, Mudassir S, et al. Participants' experiences and satisfaction with sun protection factor (SPF) 100 sunscreen in actual use conditions. Cureus. 2024;16(5):e61212. https://doi.org/10.7759/cureus.61212.
49. Nicholson A, Doherty L, Harrison A, Dixon H. Qualitative findings on consumer awareness of recommendations for effective sunscreen application. Aust N Z J Public Health. 2025;100253. https://doi.org/10.1016/j.anzjph.2025.100253.
50. D'Souza C, Kramadhari N, Skalkos E, Dutton T, Bailey J. Sun safety knowledge, practices and attitudes in rural Australian farmers: a cross-sectional study in Western New South Wales. BMC Public Health. 2021;21(1):731. https://doi.org/10.1186/s12889-021-10777-x.
51. Yaipen C, Peralta E, Rojas H, Vivar A, Herrera A, Díaz C. Conocimiento y actitudes de Medidas de protección solar En trabajadores agrícolas Tumán mayo-junio 2014. Rev Cuerpo med HNAAA. 2015;8(4):207–11.
52. Fernández-Morano T, De Troya-Martín M, Rivas-Ruiz F, Blázquez-Sánchez N, Buendía-Eisman A. Sensitivity to change of the Beach Questionnaire to behaviour, attitudes and knowledge related to sun exposure: quasi-experimental before-after study. BMC Public Health. 2015;15:60. https://doi.org/10.1186/s12889-015-1415-0.
53. Lee A, Garbutcheon-Singh KB, Dixit S, Brown P, Smith SD. The influence of age and gender in knowledge, behaviors and attitudes towards sun protection: a cross-sectional survey of Australian outpatient clinic attendees. Am J Clin Dermatol. 2015;16(1):47–54. https://doi.org/10.1007/s40257-014-0106-4.
54. Trenerry C, Fletcher C, Wilson C, Gunn K. She'll be right, mate: a mixed methods analysis of skin cancer pre-vention practices among Australian farmers-an at-risk group. Int J Environ Res Public Health. 2022;19(5):2940. https://doi.org/10.3390/ijerph19052940.
55. Buchanan N, Berktold J, Holman DM, Stein K, Prempeh A, Yerkes A. Skin cancer knowledge, awareness, beliefs and preventive behaviors among black

and Hispanic men and women. Prev Med Rep. 2018;12:203–9. https://doi.org/10.1016/j.pmedr.2018.09.017.
56. Littlewood Z, Greenfield S. Parents' knowledge, attitudes and beliefs regarding sun protection in children: a qualitative study. BMC Public Health. 2018;18(1):207. https://doi.org/10.1186/s12889-018-5091-8.
57. Calderón TA, Bleakley A, Jordan AB, Lazovich D, Glanz K. Correlates of sun protection behaviors in Racially and ethnically diverse U.S. adults. Prev Med Rep. 2018;13:346–53. https://doi.org/10.1016/j.pmedr.2018.12.006.
58. Ware OR, Dawson JE, Shinohara MM, Taylor SC. Racial limitations of Fitzpatrick skin type. Cutis. 2020;105(2):77–80.
59. Wright CY, Wilkes M, du Plessis JL, Reeder AI. Self-reported skin colour and erythemal sensitivity vs. objectively measured constitutive skin colour in an African population with predominantly dark skin. Photodermatol Photoimmunol Photomed. 2015;31(6):315–24. https://doi.org/10.1111/phpp.12191.
60. Reeder AI, Hammond VA, Gray AR. Questionnaire items to assess skin color and erythemal sensitivity: reliability, validity, and the dark shift. Cancer Epidemiol Biomarkers Prev. 2010;19(5):1167–73. https://doi.org/10.1158/1055-9965.EPI-09-1300.
61. Pimentel MA. Comment on racial limitations of Fitzpatrick skin type. Cutis. 2020;105(6):319–20. https://doi.org/10.12788/cutis.0008.
62. Eilers S, Bach DQ, Gaber R, Blatt H, Guevara Y, Nitsche K, Kundu RV, Robinson JK. Accuracy of self-report in assessing Fitzpatrick skin phototypes I through VI. JAMA Dermatol. 2013;149(11):1289–94. https://doi.org/10.1001/jamadermatol.2013.6101.
63. Buller DB, Andersen PA, Walkosz BJ, Scott MD, Beck L, Cutter GR. Rationale, design, samples, and baseline sun protection in a randomized trial on a skin cancer prevention intervention in resort environments. Contemp Clin Trials. 2016;46:67–76. https://doi.org/10.1016/j.cct.2015.11.015.
64. Panahi H, Salehi L. Perspectives of paddy workers regarding the use of sunscreen: a theory-based qualitative research. Arch Public Health. 2019;77:35. https://doi.org/10.1186/s13690-019-0361-1.
65. Lizer SK, Petrea RE. Health and Safety Needs of Older Farmers: Part I. Work Habits and Health Status. AAOHN J. 2007;55(12):485–91. https://doi.org/10.1177/216507990705501202.
66. Elliot V, Hagel L, Dosman JA, Rana M, Lawson J, Marlenga B, Trask C, Pickett W. Resilience of farm women working the third shift. J Agromedicine. 2018;23(1):70–7. https://doi.org/10.1080/1059924X.2017.1384421.
67. Asil EU, Dagli A, Aygun O. Sun protection knowledge and behaviors of agricultural workers in Turkey: a cross-sectional study. BMC Public Health. 2024;24:2571. https://doi.org/10.1186/s12889-024-20121-8.
68. Turner J, Parisi AV. Ultraviolet radiation albedo and reflectance in review: the influence to ultraviolet exposure in occupational settings. Int J Environ Res Public Health. 2018;15(7):1507. https://doi.org/10.3390/ijerph15071507.
69. Davin EL, Seneviratne SI, Ciais P, Olioso A, Wang T. Preferential cooling of hot extremes from cropland albedo management. Proc Natl Acad Sci U S A. 2014;111(27):9757–61. https://doi.org/10.1073/pnas.1317323111.
70. Sieber P, Ericsson N, Hammar T, Hansson PA. Albedo impacts of current agricultural land use: crop-specific albedo from MODIS data and inclusion in LCA of crop production. Sci Total Environ. 2022;835:155455. https://doi.org/10.1016/j.scitotenv.2022.155455.
71. Salvatierra-Rojas A, Torres-Toledo V, Müller J. Influence of surface reflection (Albedo) in simulating the sun drying of paddy rice. Appl Sci. 2020;10(15):5092. https://doi.org/10.3390/app10155092.
72. Carley A, Stratman E. Skin cancer beliefs, knowledge, and prevention practices: a comparison of farmers and nonfarmers in a midwestern population. J Agromedicine. 2025;20(2):85–94. https://doi.org/10.1080/1059924X.2015.1010059.
73. Chen LJ, Chang YJ, Shieh CF, Yu JH, Yang MC. Relationship between practices of eye protection against solar ultraviolet radiation and cataract in a rural area. PLoS One. 2021;16(7):e0255136. https://doi.org/10.1371/journal.pone.0255136.
74. Tigchelaar M, Battisti DS, Spector JT. Work adaptations insufficient to address growing heat risk for U.S. agricultural workers. Environ Res Lett. 2020;15(9):094035. https://doi.org/10.1088/1748-9326/ab86f4.
75. Berry EG, Bezecny J, Acton M, Sulmonetti TP, Anderson DM, Beckham HW, et al. Slip versus slop: a head-to-head comparison of UV-protective clothing to sunscreen. Cancers (Basel). 2022;14(3):542. https://doi.org/10.3390/cancers14030542.
76. Toro-Huamanchumo CJ, Burgos-Muñoz SJ, Vargas-Tineo LM, Perez-Fernandez J, Vargas-Tineo OW, Burgos-Muñoz RM, et al. Awareness, behavior and attitudes concerning sun exposure among beachgoers in the northern coast of Peru. PeerJ. 2019;7:e6189. https://doi.org/10.7717/peerj.6189.
77. Seité S, Del Marmol V, Moyal D, Friedman AJ. Public primary and secondary skin cancer prevention, perceptions and knowledge: an international cross-sectional survey. J Eur Acad Dermatol Venereol. 2017;31(5):815–20. https://doi.org/10.1111/jdv.14104.
78. Cambil-Martín J, López JDS, Rodríguez-Martínez A, Rivas-Ruiz F, Salazar-Granizo YE, Herrera-Molina AS, et al. Sun exposure practices, attitudes and knowledge among students and teachers at a University School of Health Sciences in Ecuador. Prev Med Rep. 2023;36:102458. https://doi.org/10.1016/j.pmedr.2023.102458.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.